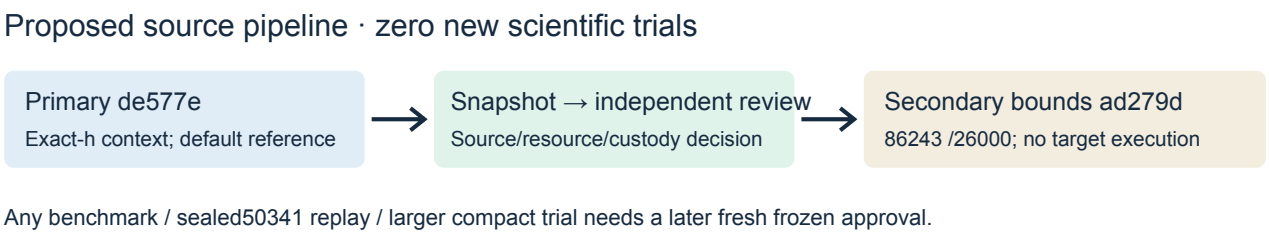
Next native source work: bounded design

GOAL-ISOU-f30a07 · source-only TASK de577e · scientific trials 0



Primary expanded-map replay optimization

From clean crypto 1180f89d568c52ac60a022358b6fd70fa9e3bd30 on fresh codex/isogeny-fixed-modulus-20261016, preserve PR1976 and frozen e82 executable. Native closure e82⟶1180 is unchanged. One immutable exact-field/modulus Newton-reciprocal context per fixed kernel, reused in selected torsion and baby/giant composition. Reference routines/default remain unchanged; opt-in Cargo feature only. No global cache or changing-divisor GCD substitution. Exact precision/prefix/residual proof plus native identity supplement required.

Controls span length 127/128/129, reduction boundaries and odd primes, interleaved contexts, selected/dense references and Horner composition. P192 / 13 and P521 / 7 cover fallback only; existing accepted P224 / 1471, kernel length 736 supplies true cached-path admission plus recomputed-hash bad-kernel/numerator negatives. No new construction/search or 50341 extraction/replay as engineering control. Real-worker positive RSS handshake and lost/zero samples before/after ack must refuse. Mandatory root/native suites and every failed attempt retained.

Compiled reduction policy goes only in public version JSON, outside embedded provenance/certificate/verdict. Companion generate accepts explicit –input SEALED_MAP_JSON; controls use bounded public fixtures. Build clean matched reference/candidate from same future commit, exact compiler/flags/lock/features and installed-control equality, hashes/version/custody. Then independent source/resource review and DEC 5cf723. No benchmark/replay target is approved now.

Future measurement contract, not a result

Frozen product/composition inputs, explicit counts and cold/amortized setup. Separate parsing/context/arithmetic/encoding/digest/verification and total OS CPU/RSS/wall. Eight fresh A/A pairs then eight ABBA blocks per case. Empirical q90 nearest-rank ceil(.9n), n=8, index 8/max; no interpolation/drop. Noise beyond 10 percent makes that metric descriptive. Full mathematical verification bytes equal by default, no dropping verdict/point fields. Any permitted provenance-only exclusion must be enumerated against actual schema before freeze.

Later one A / one B sealed 50341 replay is diagnostic under 3600 s / 8 GiB; a censored denominator gives no point speedup. Repeated whole-replay measurements needed for a speed claim. No larger expanded construction before optimized accepted replay and review. Source-level reciprocal repetition is a candidate, not a measured bottleneck.

Secondary and deferred

After primary source gate: compact envelope 86243 / 26000 digits, independently in both parsers, all math/encoding/resource gates unchanged. Parser-only edge controls, low-degree native fixtures and fresh source/build/custody review precede any later target approval. DEC 7a0aed is prospective. LL decimal conversion/digest is a deferred source lead, not a hotspot claim. Both tracks retain all old failures and category separation;54 expanded boards remain primary. Full exact paths/control grid/proof/accounting: report.md and FIXED-MODULUS-PLAN.md.